

B.Tech Degree VI Semester Examination April 2012

ME 606 CAD/CAM

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Discuss role of computers in design process.
(b) Explain various types of automated transfer machines used in production industry.
(c) With various advantages and disadvantages explain open loop and closed loop control system used in CNC machine tools.
(d) Explain various formats in manual part programming technique used in CNC machine tool control.
(e) Explain static and dynamic errors in CNC machining centers.
(f) Describe accuracy, repeatability and resolution of a CNC system.
(g) Explain P.T.P and C.P systems used in robotic control systems.
(h) Explain CIM with the help of a block diagram. Also explain its features in a production industry.

PART B

(4 x 15 = 60)

- II. (a) Explain various geometric modelling techniques used in CAD. (7)
(b) With the help of a flow chart explain various steps in developing a finite element analysis. (8)
- OR**
- III. (a) Briefly explain various data exchange protocols used in CAD/CAM. (5)
(b) Explain various levels of automation in detail. (10)
- IV. Explain with the help of neat sketches various classification of CNC machine tools in detail. (15)
- OR**
- V. With the help of a flow chart explain fundamentals of part programming. (15)
- VI. (a) Briefly discuss special design features to match machine tools to NC systems. (7)
(b) Describe the functions of automatic tool changers and tool monitoring systems used in CNC machine tools. (8)
- OR**
- VII. (a) What are the additional features of a CNC machine tool over an NC machine tool? Explain. (10)
(b) Explain various features of automatic pallet changers in detail. (5)
- VIII. (a) Explain with neat sketches the classifications of Robot according to its physical configuration in detail. (8)
(b) What are the various programming techniques used in a Robotic system? Explain. (7)
- OR**
- IX. (a) Explain various internal and external sensors used in Robot to perform its given commands. (10)
(b) Explain levels of flexibility in an FMS system in detail. (5)